SECTION A – 2 MARKS

Q1. (a)

Statistics is a branch of applied mathematics that deals with the collection, description, analysis and inference of conclusions from quantitative data. According to Croxton and Cowden, "Statistics may be defined as the collection, presentation, analysis and interpretation of numerical data."

Q1. (b)

Data collection is the process of gathering data from surveys or other sources. Data are of two types:

1. **Primary data:** Data that have not been previously published, i.e. data derived from a new or original research study and collected at the source.
2. **Secondary data:** Second-hand information which has already been collected and analysed by someone else for some purpose, e.g. the Economic Survey of India.

Q1. (c)

Tabulation is a logical and systematic presentation of statistical data in rows and columns. Rows are horizontal and columns are vertical arrangements of the data. It gives the data an orderly form, highlights their main characteristics and makes comparison easy.

Q1. (d)

1. The sum of the deviations of the items from the arithmetic mean is always zero: $\Sigma(x - \bar{x}) = 0$.
2. The sum of the squares of the deviations from the arithmetic mean is the least: $\Sigma(x - \bar{x})^2$ is a minimum.

Q1. (e)

Largest value L = 100, smallest value S = 20.
Range = L − S = 100 − 20 = **80**.
Coefficient of range = (L − S) / (L + S) = 80 / 120 = **0.67**.

Q1. (f)

Standard deviation is the square root of the arithmetic mean of the squared deviations of the items from their arithmetic mean. It is denoted by σ: $\sigma = \sqrt{(\Sigma(x - \bar{x})^2 / N)}$. It is the most widely used measure of dispersion.

Q1. (g)

Correlation is a statistical technique used to analyse the strength and direction of the relationship between two variables, e.g. income and expenditure, or height and weight. According to A.M. Tuttle, "Correlation is an analysis of the co-variation between two or more variables."

Q1. (h)

Regression is the statistical technique that expresses a functional relationship between two variables in the form of an equation, to estimate (or predict) the value of one variable from the given value of the other. The word means 'stepping back'; it was first used by Sir Francis Galton.

Q1. (i)

Classification is the process of dividing the data into different groups or classes which are homogeneous within but heterogeneous between themselves, i.e. arranging data into homogeneous groups or classes according to some common characteristics.

Q1. (j)

A **histogram** is a graph of a frequency distribution in which the class intervals are shown on the X-axis and the frequencies on the Y-axis. Each class is shown by a rectangle whose height is its frequency, and the rectangles are drawn next to one another.

SECTION B – 5 MARKS

Q2.

Classification is the process of dividing data into groups or classes which are homogeneous within but heterogeneous between themselves. There are four bases (modes) of classification:

1. **Qualitative base:** Data are classified according to some quality or attribute, such as sex, religion, literacy or intelligence. It is of two kinds: (i) **Simple classification**, when only one attribute is used, e.g. population into male and female. (ii) **Manifold classification**, when more than one attribute is used, e.g. population into male and female, then each into married and unmarried, then each into literate and illiterate.
2. **Quantitative base:** Data are classified by characteristics that can be measured, such as heights, weights, ages or income.
3. **Geographical base:** Data are classified by region or location, such as states, provinces, cities or countries, e.g. sales of a firm in Karnataka, Tamil Nadu, Andhra Pradesh and Kerala.
4. **Chronological (temporal) base:** Data are classified by their time of occurrence, such as years, months, weeks or days, e.g. sales of a firm in 2012, 2013, 2014 and 2015.

Q3.

Primary data are collected by the following methods:

1. **Direct personal investigation:** The information is collected by direct personal interviews, so it is first-hand and original. It is useful when the scope of the enquiry is small. Response is good and the data are accurate, but it is expensive, time-consuming and open to personal bias.
2. **Indirect oral interviews:** Information is collected not from the source directly but by interviewing persons closely related to the problem, as in cases of theft or murder. It saves time and cost and covers a vast area, but third parties may not co-operate.
3. **Mailed questionnaire method:** A questionnaire (a set of questions on the enquiry) is mailed to the informants with a request to answer and return it within a specified time. The questions should be short, clear, few in number, simple and impersonal.
4. **Interview schedule method:** When informants are uneducated or do not respond, enumerators appointed by the investigator meet them with interview schedules, ask the questions and record the answers. Success depends on the sincerity and efficiency of the enumerators.
5. **From local agents:** Local agents or correspondents, appointed by the sponsoring authorities and familiar with local language and conditions, send the information. It is used by government agencies, newspapers and periodicals.

Primary data are also collected **through telephone**, which is quick and gives accurate information.

Q4.

x	f	cf
20	25	25
60	48	73
80	60	133
100	57	190
120	29	219
140	10	229
160	8	237
200	2	239
Total	**N = 239**	

Q1 = size of (N + 1)/4 th item = 240/4 = 60th item. The 60th item lies in cf 73, so **Q1 = 60**.
Q3 = size of 3(N + 1)/4 th item = 3 × 60 = 180th item. The 180th item lies in cf 190, so **Q3 = 100**.
Quartile deviation = (Q3 − Q1) / 2 = (100 − 60) / 2 = **20**.
Coefficient of QD = (Q3 − Q1) / (Q3 + Q1) = 40 / 160 = **0.25**.

Q5.

x	log x
20	1.3010
350	2.5441
450	2.6532
600	2.7782
861	2.9350
958	2.9814
N = 6	**Σ log x = 15.1929**

GM = Antilog (Σ log x / N) = Antilog (15.1929 / 6) = Antilog 2.5322

GM = 340.5

Q6.

Title: Distribution of employees in an office according to sex, salary grade and designation

Designation	Sex	Below 3,000	3,000 to 5,000	5,000 and above	Total
Supervisor	Male				
	Female				
	Total				
Assistant	Male				
	Female				
	Total				
Clerk	Male				
	Female				
	Total				
Grand total					

The column headings show the salary grade (Rs.), the stub shows designation and sex, and each cell will hold the number of employees.

Source: ..... **Note:** .....

Q7.

N = 20, $\bar{x}$ = 4, $\bar{y}$ = 2, Σx² = 1,680, Σy² = 320, Σxy = 480.

1. **Variance of x:** $\sigma x^2 = \Sigma x^2/N - \bar{x}^2 = 1{,}680/20 - 4^2 = 84 - 16 = 68$.
2. **Variance of y:** $\sigma y^2 = \Sigma y^2/N - \bar{y}^2 = 320/20 - 2^2 = 16 - 4 = 12$.
3. **Covariance:** $\Sigma xy/N - \bar{x}\bar{y} = 480/20 - (4 \times 2) = 24 - 8 = 16$.
4. **Regression coefficients:** bxy = 16/12 = **1.333**; byx = 16/68 = **0.235**.

Regression equation of x on y: $x - \bar{x} = bxy\,(y - \bar{y})$

$x - 4 = 1.333\,(y - 2)$, so **x = 1.333y + 1.333**

Regression equation of y on x: $y - \bar{y} = byx\,(x - \bar{x})$

$y - 2 = 0.235\,(x - 4)$, so **y = 0.235x + 1.06**

When y = 7: x = 1.333 × 7 + 1.333 = **10.67**

SECTION C – 10 MARKS

Q10.

The classes are inclusive, so they are changed to exclusive classes (9.5 – 19.5 and so on) for the median and the mode.

Class	Exclusive class	f	Mid-point (x)	fx	cf
10 – 19	9.5 – 19.5	10	14.5	145.0	10
20 – 29	19.5 – 29.5	18	24.5	441.0	28
30 – 39	29.5 – 39.5	8	34.5	276.0	36
40 – 49	39.5 – 49.5	6	44.5	267.0	42
50 – 59	49.5 – 59.5	4	54.5	218.0	46
60 – 69	59.5 – 69.5	2	64.5	129.0	48
70 – 79	69.5 – 79.5	5	74.5	372.5	53
80 – 89	79.5 – 89.5	7	84.5	591.5	60
Total		**N = 60**		**Σfx = 2,440**	

Mean

$\bar{x} = \Sigma fx / N = 2{,}440 / 60 =$ **40.67**

Median

N/2 = 60/2 = 30th item, which lies in the class 29.5 – 39.5 (cf 36).

L = 29.5, cf = 28, f = 8, i = 10

Median = L + ((N/2 − cf) / f) × i = 29.5 + ((30 − 28) / 8) × 10 = 29.5 + 2.5 = **32**

Mode

The highest frequency is 18, so the modal class is 19.5 – 29.5.

L = 19.5, f1 = 18, f0 = 10, f2 = 8, i = 10

Mode = L + ((f1 − f0) / (2f1 − f0 − f2)) × i = 19.5 + (8 / (36 − 10 − 8)) × 10 = 19.5 + (8/18) × 10 = 19.5 + 4.44 = **23.94**

Mean = 40.67, Median = 32, Mode = 23.94

Q11.

Assumed mean A = 35, class interval i = 10, d = (m − 35) / 10.

Class	f	Mid-point (m)	d	fd	fd²
10 – 20	10	15	−2	−20	40
20 – 30	18	25	−1	−18	18
30 – 40	20	35	0	0	0
40 – 50	6	45	1	6	6
50 – 60	4	55	2	8	16
60 – 70	2	65	3	6	18
Total	**N = 60**			**Σfd = −18**	**Σfd² = 98**

Mean

$\bar{x} = A + (\Sigma fd / N) \times i = 35 + (-18 / 60) \times 10 = 35 - 3 =$ **32**

Standard deviation

$\sigma = \sqrt{(\Sigma fd^2/N - (\Sigma fd/N)^2)} \times i = \sqrt{(98/60 - (-18/60)^2)} \times 10$

$\sigma = \sqrt{(1.6333 - 0.09)} \times 10 = \sqrt{1.5433} \times 10 = 1.2423 \times 10 =$ **12.42**

Coefficient of variation

CV = (σ / $\bar{x}$) × 100 = (12.42 / 32) × 100 = **38.82%**

Q12.

x	y	x²	y²	xy
8	25	64	625	200
10	30	100	900	300
15	32	225	1,024	480
17	35	289	1,225	595
20	37	400	1,369	740
22	40	484	1,600	880
24	42	576	1,764	1,008
25	45	625	2,025	1,125
Σx = 141	**Σy = 286**	**Σx² = 2,763**	**Σy² = 10,532**	**Σxy = 5,328**

N = 8

Coefficient of correlation

$r = (N\Sigma xy - \Sigma x\Sigma y) / \sqrt{[(N\Sigma x^2 - (\Sigma x)^2)(N\Sigma y^2 - (\Sigma y)^2)]}$

$N\Sigma xy - \Sigma x\Sigma y = 8 \times 5{,}328 - 141 \times 286 = 42{,}624 - 40{,}326 = 2{,}298$

$N\Sigma x^2 - (\Sigma x)^2 = 8 \times 2{,}763 - 141^2 = 22{,}104 - 19{,}881 = 2{,}223$

$N\Sigma y^2 - (\Sigma y)^2 = 8 \times 10{,}532 - 286^2 = 84{,}256 - 81{,}796 = 2{,}460$

$r = 2{,}298 / \sqrt{(2{,}223 \times 2{,}460)} = 2{,}298 / \sqrt{54{,}68{,}580} = 2{,}298 / 2{,}338.5 =$ **0.983**

Probable error

$PE = 0.6745 \times (1 - r^2) / \sqrt{N} = 0.6745 \times (1 - 0.9657) / \sqrt{8} = 0.6745 \times 0.0343 / 2.828 =$ **0.0082**

r = 0.983, PE = 0.0082. There is a very high positive correlation between x and y. Since r is more than 6 times the PE (6 × 0.0082 = 0.049), the correlation is significant.